Azure AI Foundry Platform Overview

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> Sample foundation document written for testing a domain-expert RAG. Azure AI Foundry evolves quickly and product names have changed over time; confirm details in the official Microsoft Learn documentation.

1. What Azure AI Foundry is

Azure AI Foundry is Microsoft's platform for building, evaluating and operating generative AI applications and agents on Azure. It brings together a model catalog, model deployments, playgrounds, prompt and agent tooling, evaluation, content safety and monitoring in one portal and SDK. It was previously known as Azure AI Studio. Azure OpenAI Service models are available through it alongside models from other providers.

2. Resource model

Foundry organises work into **projects**. A project is where a team keeps its deployments, agents, data, evaluations and connections.

Concept	What it is	Notes
Foundry resource and project	The current default: projects created on an Azure AI Foundry resource	Supports the agent service, models and evaluations with simpler setup
Hub	A shared resource that groups hub-based projects and centralises networking, compute, storage and connections	Used for features that need hub-based projects, such as some prompt flow and managed compute scenarios
Hub-based project	A project that inherits security and connections from its hub	Isolates each team's work while sharing governance
Connection	A stored link to another service, such as Azure AI Search, storage or an external API	Prefer Microsoft Entra ID (managed identity) over keys

Hub versus project, in short: the hub is the administrative and security boundary set

up by the platform team (network isolation, shared compute, storage and key vault, shared connections); projects are the workspaces where developers build. Many teams can have projects under one hub without seeing each other's work.

3. Identity and access

Use Microsoft Entra ID for users and managed identities for services. Azure role-based access control (RBAC) governs who can create deployments, run evaluations or manage the resource; built-in roles include owner and contributor roles for administration and developer or user roles for building within a project. Grant developers project-level roles rather than subscription-wide ones. Disable key-based authentication where possible and let applications call models with their managed identity.

4. Networking

For production, disable public network access and use private endpoints so traffic to the Foundry resource, Azure OpenAI, Azure AI Search and storage stays on your virtual network. Hubs can use a managed virtual network with outbound rules that allow only approved destinations. Combine with Azure Firewall or network security groups for egress control and with Azure Policy to enforce these settings across subscriptions.

5. The model catalog

The model catalog lists models from Microsoft and Azure OpenAI and from providers such as Meta (Llama), Mistral, Cohere and others, plus Microsoft's small Phi models. Each model card shows capabilities, benchmarks, licence and supported deployment options. Model benchmarks and leaderboards in the portal help compare quality, cost and throughput before choosing.

6. Deployment options

Deployment type	How it works	Choose when
Standard	Pay per token; processing in the resource's region	Development and moderate production traffic with regional data processing

Global Standard	Pay per token; requests routed across Azure's global capacity	Highest default quota and availability when data may be processed in any region
Data Zone Standard	Pay per token within a geographic zone such as the EU or US	Data residency within a zone with better capacity than single-region
Provisioned (PTU)	Reserved throughput units billed hourly or by reservation	Predictable high volume and consistent latency
Batch	Asynchronous jobs at a lower price, returned within a target window	Large offline workloads such as document enrichment
Serverless API (models as a service)	Pay-per-token hosting of catalog models from other providers	Open or partner models without managing compute
Managed compute	Dedicated virtual machines host the model	Custom or fine-tuned open models needing full control

Choosing for an Azure OpenAI model: start with Global Standard for the best availability if your compliance rules allow processing anywhere; use Data Zone or Standard when data must stay in a zone or region; move to Provisioned when traffic is steady and you need guaranteed throughput and predictable latency.

7. Quotas

Pay-per-token deployments have quotas expressed in tokens per minute (TPM) and requests per minute (RPM) per model and region. Assign quota across deployments deliberately, handle HTTP 429 responses with retry and backoff, and monitor usage in Azure Monitor.

8. SDKs and APIs

Applications use the Azure AI Foundry SDK (for example the `azure-ai-projects` package for projects, agents, evaluations and connections, and model inference clients) or the OpenAI-compatible Azure OpenAI endpoints. Authenticate with `DefaultAzureCredential` so the same code uses a developer's login locally and a managed identity in Azure.

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient

project = AIProjectClient(endpoint=PROJECT_ENDPOINT, credential=DefaultAzureCred
```

